THE CANONICAL HEIGHT AND THE REGULATOR, WITH SOLUTIONS

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In this worksheet we see how the canonical height $\hat{h}$ on an elliptic curve E allows us to think about the Mordell–Weil group $E(\mathbb{Q})$ as a lattice Λ in an euclidean space V. As a nice application of this, we have Néron's count of the rational points of bounded height. We use `Magma` to double check our computations.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by a Weierstrass equation.
- The height $h_x(P) \coloneqq \log H(x(P))$ of a rational point $P \neq O$, where $H(u/w) \coloneqq \max(|u|, |w|)$ for u/w in lowest terms, and $h_x(O) \coloneqq 0$. `Magma` calls it `NaiveHeight`.
- The canonical height $\hat{h}(P) \coloneqq \frac{1}{2} \lim_{n\to\infty} 4^{-n} h_x(2^n P)$ of a rational point P, and the height pairing $\langle P, Q\rangle \coloneqq \hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q)$, so that $\langle P, P\rangle = 2\hat{h}(P)$ [Sil09, §VIII.9]. The limit exists by Problem 1.5(2). `Magma` computes them as `CanonicalHeight(P)`, which is $2\hat{h}(P)$, not $\hat{h}(P)$, and `HeightPairing(P, Q)`.

1. Tate's construction of the canonical height

This section is Tate's construction of the canonical height [Sil09, Proposition VIII.9.1], first on an abelian group, as in [HS00, Theorem B.4.1], then on the rational points of an elliptic curve. Problem 1.5 asks for the proofs, which need only the following ingredients.

Definition 1.1 (Quadratic heights)**.** Let A be an abelian group. A height on A is a function $h\colon A \to \mathbb{R}$. Two heights satisfy $h = h' + O(1)$ if $|h - h'|$ is bounded on A. A height h is quasi-quadratic if

$$h(P+Q) + h(P-Q) = 2h(P) + 2h(Q) + O(1),$$

that is, if the difference of the two sides is bounded on $A \times A$. It is quadratic if the two sides are equal for all $P, Q \in A$ (the parallelogram law).

Definition 1.2 (Heights on an elliptic curve [Sil09, §VIII.6])**.** Let K be a number field, h the absolute logarithmic height on $\mathbb{P}^1(K)$ [Sil09, §VIII.5], E an elliptic curve over K, and $f \in K(E)$ a nonconstant function. The height relative to f is the height $h_f(P) \coloneqq h(f(P))$ on $E(K)$. The function f is even if $f(-P) = f(P)$ for all P, as x is. Over $\mathbb{Q}$, h_x is the height of the Standing notation.

Theorem 1.3 (Quasi-parallelogram law [Sil09, Theorem VIII.6.2])**.** *In Definition 1.2, if f is even, then h_f is a quasi-quadratic height on $E(K)$.*

Lemma 1.4 (Comparison of heights [Sil09, Lemma VIII.6.3])**.** *In Definition 1.2, let $f, g \in K(E)$ be nonconstant even functions. Then $(\deg g)\, h_f = (\deg f)\, h_g + O(1)$ on $E(K)$.*

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Problem 1.5 (Tate's construction)**.**

(1) Let A be an abelian group, and h a quasi-quadratic height on A. Prove that for every $P \in A$ the limit

$$\hat{h}(P) \coloneqq \lim_{n\to\infty} 4^{-n} h(2^n P)$$

exists, and that $\hat{h}$ is the unique quadratic height on A with $\hat{h} = h + O(1)$.

Solution. **Telescoping gives a Cauchy sequence. A bounded quadratic height is 0.** Let $c \geqslant 0$ bound the difference of the two sides of the quasi-parallelogram law. With $Q = P$ it gives $|h(2P) + h(0) - 4h(P)| \leqslant c$, so $|h(2P) - 4h(P)| \leqslant C$ for all P, with $C \coloneqq c + |h(0)|$. Apply this to $2^n P$, divide by 4^{n+1}, and sum over $M \leqslant n < N$:

$$\begin{aligned} \left|4^{-N} h(2^N P) - 4^{-M} h(2^M P)\right| &\leqslant \sum_{n=M}^{N-1} 4^{-(n+1)} \left|h(2^{n+1}P) - 4h(2^n P)\right| \\ &\leqslant C \sum_{n\geqslant M} 4^{-(n+1)} = \tfrac{1}{3} C\, 4^{-M}. \end{aligned}$$

So the sequence is Cauchy, and $\hat{h}(P)$ exists. With $M = 0$ and $N \to \infty$, $|\hat{h}(P) - h(P)| \leqslant C/3$. The quasi-parallelogram law for $2^n P$ and $2^n Q$, divided by 4^n, shows that the two sides of the parallelogram law for $4^{-n} h(2^n \cdot)$ differ by at most $c\, 4^{-n}$. In the limit they are equal, so $\hat{h}$ is quadratic.

For uniqueness, let q and q' be quadratic heights with $q = q' + O(1)$. Their difference $d \coloneqq q - q'$ is quadratic and bounded. The parallelogram law for d with $P = Q = 0$ gives $d(0) = 0$, and with $Q = P$ it gives $d(2P) = 4d(P)$. So $d(P) = 4^{-n} d(2^n P)$ for every n, which tends to 0 since d is bounded. Hence $q = q'$.

(2) Let K, E and f be as in Definition 1.2, with f even. Deduce that for every $P \in E(K)$ the limit

$$\frac{1}{\deg f} \lim_{N\to\infty} 4^{-N} h_f(2^N P)$$

exists, that it is a quadratic height on $E(K)$, and that it does not depend on f.

Solution. **Apply (1) to $h_f/\deg f$. Uniqueness compares two choices of f.** By Theorem 1.3, $h_f/\deg f$ is a quasi-quadratic height on $E(K)$. By (1), the limit exists and is the unique quadratic height $\hat{h}_f$ with $\hat{h}_f = h_f/\deg f + O(1)$. Let $g \in K(E)$ be another nonconstant even function. By Lemma 1.4, $h_f/\deg f = h_g/\deg g + O(1)$. So $\hat{h}_f$ and $\hat{h}_g$ are quadratic heights equivalent to $h_g/\deg g$, and $\hat{h}_f = \hat{h}_g$ by the uniqueness in (1). Over $\mathbb{Q}$, with $f = x$ of degree 2, this is the canonical height $\hat{h}$ of the Standing notation.

Remark. Silverman proves the existence and the independence of f directly, by the same telescoping sum [Sil09, Proposition VIII.9.1], and the parallelogram law afterwards [Sil09, Theorem VIII.9.3]. Hindry and Silverman do the abstract construction for any map ϕ with $h \circ \phi = \alpha h + O(1)$ and $\alpha > 1$ [HS00, Theorem B.4.1].

2. The lattice of rational points

This section draws [Sil09, Proposition VIII.9.6] for the curve $y^2 = x^3 + 17$ of [Sil09, Example III.2.4], after [HS00, §B.5]. The rational points P_1 and P_3 of Problem 2.5 carry Silverman's names. He notes that they generate the Mordell–Weil group, and that this is true but not so easy to prove. Problem 2.5 proves it with `Magma`'s `CPSHeightBounds`, which returns real numbers l and u

with $l \leqslant h_x(P) - 2\hat{h}(P) \leqslant u$ for every rational point P [BCP97]. First, two facts about lattices in general.

Definition 2.1 (Lattices and covolume [HS00, §B.5])**.** A lattice in a real vector space V of dimension r is a subgroup Λ generated by a basis $\lambda_1, \dots, \lambda_r$ of V. If V is Euclidean, the covolume $\operatorname{covol}(\Lambda)$ is the volume of $\{t_1\lambda_1 + \cdots + t_r\lambda_r : 0 \leqslant t_i < 1\}$.

Proposition 2.2 (The covolume [HS00, Exercise B.14(b)])**.** *In Definition 2.1,* $\operatorname{covol}(\Lambda)^2 = \det(\lambda_i \cdot \lambda_j)_{i,j}$.

From now on, V and Λ denote the following.

Proposition 2.3 (The height lattice [Sil09, Proposition VIII.9.6])**.** *Let* $V := E(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{R}$*, let* $v_P := P \otimes 1$ *for a rational point* P*, and let* Λ *be the image of* $E(\mathbb{Q})$ *in* V*. The pairing extends to an inner product on* V*, with* $|v_P|^2 = 2\hat{h}(P)$*. Then* Λ *is a lattice in* V*, and the kernel of* $E(\mathbb{Q}) \to \Lambda$ *is* $E(\mathbb{Q})_{\mathrm{tors}}$.

Definition 2.4 (The elliptic regulator [Sil09, §VIII.9])**.** Let r be the rank of E, and $Q_1, \dots, Q_r$ rational points whose images form a basis of Λ. The elliptic regulator of E is $R_E := \det\big(\langle Q_i, Q_j\rangle\big)_{1\leqslant i,j\leqslant r}$. By Proposition 2.2, $\sqrt{R_E} = \operatorname{covol}(\Lambda)$ is the covolume of Λ in V.

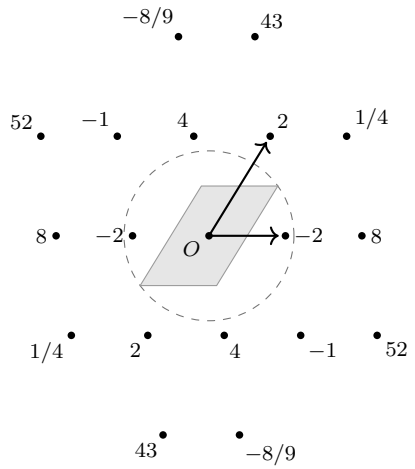


FIGURE 1. The rational points of the curve E of Problem 2.5 near the origin of the plane V. Each dot is the vector v_P of a rational point $P = aP_1 + bP_3$ with $2\hat{h}(P) \leqslant 3.4$, labeled by its x-coordinate. Since $|v_P|^2 = 2\hat{h}(P)$, the labels grow more complicated outward: integers near O, and the first denominators, $1/4$ and $-8/9$, farther out. The rational points P and $-P$ have the same x-coordinate and sit opposite each other. The arrows are v_{P_1} and v_{P_3}. The shaded parallelogram $\{sv_{P_1} + tv_{P_3} : |s|, |t| \leqslant \frac{1}{2}\}$ has area $\sqrt{\det G} = 0.592076$, and every vector of V is a vector of $\mathbb{Z}v_{P_1} + \mathbb{Z}v_{P_3}$ plus a vector in it. The dashed circle $2\hat{h} = 0.561255$ passes through its two farthest corners $\pm\frac{1}{2}(v_{P_1} + v_{P_3})$, so it contains the parallelogram.

Problem 2.5 (Two rational points generate $E(\mathbb{Q})$)**.**
Let $E\colon y^2 = x^3 + 17$ over $\mathbb{Q}$, with LMFDB label `10404.f2`, and let $P_1 = (-2, 3)$ and $P_3 = (2, 5)$ be two of its rational points. Let $g_{ij} := \langle P_i, P_j\rangle$ for $i, j \in \{1, 3\}$, and let $G := (g_{ij})$ be the Gram matrix of P_1 and P_3. `Magma` computes the rank and the torsion of E, the matrix G and

its determinant, the bounds l and u, and the x-coordinates of the rational points $P \neq O$ with $H(x(P)) \leqslant 5$:

```
R := RealField(6);
E := EllipticCurve([0, 0, 0, 0, 17]);
Rank(E); #TorsionSubgroup(E);                    // 2 true
                                                 // 1
P1 := E![-2, 3];  P3 := E![2, 5];
G := HeightPairingMatrix([P1, P3]);  ChangeRing(G, R);
// [0.454617 0.363961]
// [0.363961 1.06248]
R!Determinant(G), R!Regulator(E);                // 0.350554 0.350554
l, u := CPSHeightBounds(E);  R!l, R!u;           // -1.63999 1.18106
X := {a/d : a in [-5..5], d in [1..5] | GCD(a, d) eq 1 and IsPoint(E, a/d)};
X;                                               // { -2, -1, 1/4, 2, 4 }
X subset {(m*P1 + n*P3)[1] : m, n in [-2..2] | [m, n] ne [0, 0]};   // true
```

The items below show that P_1 and P_3 generate $E(\mathbb{Q})$.

(1) Show that v_{P_1} and v_{P_3} form a basis of V.

SOLUTION. **$\dim V = 2$, and $\det G \neq 0$.** The rank of E is 2, by the Magma computation, so V has dimension 2. A linear relation $\alpha v_{P_1} + \beta v_{P_3} = 0$ with $(\alpha, \beta) \neq (0, 0)$ would give $(\alpha, \beta)\, G = 0$, by pairing it with v_{P_1} and v_{P_3}. But $\det G = 0.350554 \neq 0$. So v_{P_1} and v_{P_3} are independent, hence a basis.

(2) Let P be a rational point. Show that there are $m, n \in \mathbb{Z}$ such that $Q := P - mP_1 - nP_3$ has v_Q in the shaded parallelogram of Fig. 1. Show that $Q \in \mathbb{Z}P_1 + \mathbb{Z}P_3$ if and only if $P \in \mathbb{Z}P_1 + \mathbb{Z}P_3$.

SOLUTION. **Take m, n nearest to the coordinates of v_P.** By (1), $v_P = s v_{P_1} + t v_{P_3}$ with $s, t \in \mathbb{R}$. Let m, n be integers nearest to s, t, so that $|s - m|, |t - n| \leqslant \frac{1}{2}$. Since $P \mapsto v_P$ is a homomorphism, $v_Q = (s - m) v_{P_1} + (t - n) v_{P_3}$, which lies in the shaded parallelogram. And $P - Q = mP_1 + nP_3$ lies in $\mathbb{Z}P_1 + \mathbb{Z}P_3$, so P and Q lie in it or not together.

(3) Show that every rational point Q with v_Q in the shaded parallelogram has

$$2\hat{h}(Q) \leqslant \tfrac{1}{4} \max(g_{11} + 2g_{13} + g_{33},\ g_{11} - 2g_{13} + g_{33}),$$

and deduce that $H(x(Q)) \leqslant 5$.

SOLUTION. **$2\hat{h}(Q) \leqslant 0.561255$, so $H(x(Q)) \leqslant e^{1.74231} = 5.71$.** The corners of the parallelogram are $\pm\frac{1}{2}(v_{P_1} + v_{P_3})$ and $\pm\frac{1}{2}(v_{P_1} - v_{P_3})$. By bilinearity, $|v_{P_1} \pm v_{P_3}|^2 = g_{11} \pm 2g_{13} + g_{33}$, which is 2.24502 and 0.789176. The convex function $|v|^2$ is largest on the parallelogram at a corner, so

$$2\hat{h}(Q) = |v_Q|^2 \leqslant \tfrac{1}{4} \max(2.24502, 0.789176) = 0.561255.$$

Then $h_x(Q) \leqslant 2\hat{h}(Q) + u \leqslant 0.561255 + 1.18106 = 1.74231$, so $H(x(Q)) \leqslant e^{1.74231} = 5.71$. Since $H(x(Q))$ is an integer, $H(x(Q)) \leqslant 5$.

(4) Conclude that P_1 and P_3 generate $E(\mathbb{Q})$.

SOLUTION. **Otherwise some $Q \notin \mathbb{Z}P_1 + \mathbb{Z}P_3$ has $H(x(Q)) \leqslant 5$, and there is none.** Suppose that a rational point P is not in $\mathbb{Z}P_1 + \mathbb{Z}P_3$. By (2) there is a rational point $Q \notin \mathbb{Z}P_1 + \mathbb{Z}P_3$ with v_Q in the shaded parallelogram. In particular $Q \neq O$, and $H(x(Q)) \leqslant 5$ by (3). The Magma search shows that $x(Q) = x(R)$ for some $R \in \mathbb{Z}P_1 + \mathbb{Z}P_3$. Then $Q = \pm R$ lies in $\mathbb{Z}P_1 + \mathbb{Z}P_3$, a contradiction.

Remark. This is the saturation step of a computation of $E(\mathbb{Q})$. `Magma`'s `Regulator(E)` runs it on its own, so its agreement with $\det G$ is a consistency check, not a proof. In general, if the images of rational points $Q_1, \dots, Q_r$ generate a subgroup of index N in Λ, then $\det(\langle Q_i, Q_j \rangle) = N^2 R_E$.

3. Néron's counting theorem

This section is Néron's counting theorem [HS00, Theorem B.6.3] for the curve of Problem 2.5, with its constant, through the count of lattice points in ellipsoids [HS00, Lemma B.6.4].

Lemma 3.1 (Lattice points in ellipsoids [HS00, Lemma B.6.4])**.** *Let r be the rank of E, suppose that $r \geqslant 1$, and let q be a positive definite quadratic form on V. Then, as $T \to \infty$,*

$$\#\{\lambda \in \Lambda : q(\lambda) \leqslant T\} = \frac{\mathrm{vol}\{v \in V : q(v) \leqslant 1\}}{\mathrm{covol}(\Lambda)}\, T^{r/2} + O\big(T^{(r-1)/2}\big).$$

Theorem 3.2 (Néron's counting theorem [HS00, Theorem B.6.3])**.** *Suppose that E has rank $r \geqslant 1$. There is a constant $c > 0$ such that, as $T \to \infty$,*

$$\#\{P \in E(\mathbb{Q}) : h_x(P) \leqslant T\} = c\, T^{r/2} + O\big(T^{(r-1)/2}\big).$$

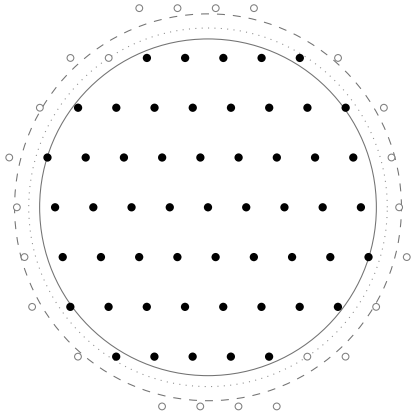


FIGURE 2. The rational points of the curve E of Problem 2.5 counted by Néron's theorem for $T = 10$. Each dot is the vector v_P of a rational point P with $|v_P|^2 = 2\hat{h}(P) \leqslant 14$. It is filled if $h_x(P) \leqslant 10$, and open otherwise. There are 53 filled dots. If h_x were equal to $2\hat{h}$, the filled dots would be exactly those inside the dotted circle $2\hat{h} = 10$. Instead, $h_x - 2\hat{h}$ is only known to lie between $l = -1.63999$ and $u = 1.18106$. So every rational point inside the solid circle $2\hat{h} = 10 - u$ has $h_x \leqslant 10$: all 49 dots there are filled. No rational point outside the dashed circle $2\hat{h} = 10 - l$ has $h_x \leqslant 10$: all dots there are open. In the ring between the two circles, the answer depends on $h_x - 2\hat{h}$, which has no pattern: 4 of its 10 dots are filled. The ring has area $\pi(u - l)$ for every T, while the disc inside it has area about πT. So, for large T, the count is about the area of the disc divided by the area $\sqrt{R_E}$ of a fundamental parallelogram.

Problem 3.3 (Counting rational points of bounded height)**.**
Let E, P_1, P_3, G, l and u be as in Problem 2.5. Continuing its listing, `Magma` counts the rational points with $h_x \leqslant T$:

```
Gi := G^-1;
count := function(T)
  N := [Floor(Sqrt((T - l)*Gi[i, i])) : i in [1, 2]];
  return #[0 : a in [-N[1]..N[1]], b in [-N[2]..N[2]] | NaiveHeight(a*P1 + b*P3) le T];
end function;
[<T, count(T)> : T in [10, 20, 50, 100]];
// [ <10, 53>, <20, 103>, <50, 263>, <100, 529> ]
```

The function `count` runs over a box that contains every $(a, b) \in \mathbb{Z}^2$ with $2\hat{h}(aP_1 + bP_3) \leqslant T - l$. The items below prove Theorem 3.2 for E, with $c = \pi/\sqrt{R_E}$.

(1) Show that $P \mapsto v_P$ is a bijection $E(\mathbb{Q}) \to \Lambda$, that $R_E = \det G$, and that $\operatorname{covol}(\Lambda) = \sqrt{R_E}$.

Solution. **$E(\mathbb{Q})_{\mathrm{tors}} = 0$, and (P_1, P_3) is a basis by Problem 2.5.** The torsion of E is trivial, by the `Magma` computation of Problem 2.5, so $P \mapsto v_P$ is injective by Proposition 2.3, and it is onto Λ by definition. By Problem 2.5, P_1 and P_3 generate $E(\mathbb{Q})$, so their images form a basis of Λ, and $R_E = \det G = 0.350554$ by Definition 2.4. Then $\operatorname{covol}(\Lambda) = \sqrt{R_E}$ by Proposition 2.2.

(2) Show that

$$\{P : 2\hat{h}(P) \leqslant T - u\} \subseteq \{P : h_x(P) \leqslant T\} \subseteq \{P : 2\hat{h}(P) \leqslant T - l\},$$

and check it against Fig. 2.

Solution. **$2\hat{h} + l \leqslant h_x \leqslant 2\hat{h} + u$.** If $2\hat{h}(P) \leqslant T - u$, then $h_x(P) \leqslant 2\hat{h}(P) + u \leqslant T$. If $h_x(P) \leqslant T$, then $2\hat{h}(P) \leqslant h_x(P) - l \leqslant T - l$. In Fig. 2, with $T = 10$, all 49 dots inside the solid circle are filled, and every dot outside the dashed circle is open.

(3) Deduce that the number of rational points P with $h_x(P) \leqslant T$ is $cT + O(\sqrt{T})$ with $c = \pi/\sqrt{R_E}$, and compare cT with the output.

Solution. **$c = \pi/\sqrt{0.350554} = 5.30606$, and $cT = 53.1, 106.1, 265.3, 530.6$.** By (1), the outer sets of (2) correspond to the lattice points $\lambda \in \Lambda$ with $|\lambda|^2 \leqslant T - u$ and $|\lambda|^2 \leqslant T - l$. Apply Lemma 3.1 with $q(v) = |v|^2$, whose unit disc has area π, and $\operatorname{covol}(\Lambda) = \sqrt{R_E}$. Both outer sets have $\pi(T + O(1))/\sqrt{R_E} + O(\sqrt{T}) = cT + O(\sqrt{T})$ elements, and so does the middle one. The counts 53, 103, 263, 529 differ from cT by less than 4.

(4) Suppose that you did not know the rank of E. How would the counts reveal it?

Solution. **Doubling T doubles the count, so $r/2 = 1$ and $r = 2$.** By Theorem 3.2, the count $N(T)$ grows like $T^{r/2}$. Here $N(20)/N(10)$, $N(100)/N(50)$ are close to 2, so $r/2 = 1$. In general $r = \lim 2 \log N(T)/\log T$: the rank can be read off from a count of rational points.

Remark. Néron stated the theorem with the multiplicative height $H_x = e^{h_x}$, where it reads $c(\log T)^{r/2}$ [HS00, Theorem B.6.3]. Rational points are sparse: on this curve, exactly 529 of them have an x-coordinate whose numerator and denominator are at most $e^{100} \approx 10^{43}$.

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